THE INCOMPLETENESS OF OBSERVATION

Why Quantum Mechanics and General Relativity Cannot Be Unified From Within

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Abstract

Physical observers are strictly subsystems of the deterministic universes they measure, permanently barred from accessing degrees of freedom beyond their causal reach. This paper proves that the structural constraints imposed by this epistemic reduction necessarily give rise to quantum mechanics. We show that an observer coupled to a slow, high-capacity hidden sector on a finite-dimensional configuration space must describe the visible sector using P-indivisible stochastic dynamics. By the stochastic-quantum correspondence, this reduction is mathematically equivalent to unitarily evolving quantum mechanics.

Crucially, the converse also holds: any quantum system, realized as a deterministic dilation, requires precisely these conditions—non-trivial coupling, slow-bath memory, and sufficient hidden-sector capacity—establishing a full equivalence between quantum mechanics and embedded observation. Consequently, the Schrödinger equation, the Born rule, and Bell inequality violations emerge as structural consequences of this equivalence, requiring no independent quantum postulates.

Applying this theorem to the cosmological horizon, which provides an exact physical realization of these causal conditions, uniquely determines the emergent action scale. Partition-relativity, dimensional analysis, and the thermodynamic consistency of the horizon fix $\hbar = c^3\epsilon^2/(4G)$, establishing the geometric discreteness scale at $\epsilon = 2\,l_p$ and recovering the exact Bekenstein–Hawking entropy formula. Furthermore, this framework reframes the cosmological constant problem from a theoretical failure into corroborating evidence. The 10^{122} discrepancy is precisely S_{dS}, representing the exact information compression ratio of the emergent quantum description. Consequently, the observed macroscopic vacuum energy is not a fine-tuned anomaly, but the mandatory classical baseline predicted by the framework. Falsifiable predictions include dark energy evolution in Running Vacuum Model form ($\Lambda_{\mathrm{eff}} = \Lambda_0 + \nu H^2$) with a coefficient $\nu_{\mathrm{OI}} = \Omega_\Lambda/(2\,\ln(c/\epsilon H_0)) \approx 2.45 \times 10^{-3}$, consistent with current observational fits, and distinct gravitational wave echoes near black hole horizons.

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PART I: THE GENERAL THEOREM

1. Introduction

1.1 The Problem of Embedded Observation

Physical observers are subsystems of the systems they measure. This elementary fact has consequences that have not been traced. An observer embedded in a deterministic universe cannot access the complete state: degrees of freedom beyond the observer's causal reach are permanently hidden, and the observer's description of the visible sector is necessarily a reduced one obtained by marginalizing over the hidden sector. The question is what structural constraints that reduction imposes on the form of the observer's physical laws.

The observer's epistemic situation has been foregrounded before. QBism treats quantum states as expressions of an agent's beliefs; relational quantum mechanics defines states relative to interacting subsystems; 't Hooft's cellular automaton interpretation [1] derives quantum behavior from deterministic dynamics through information loss. These programs take observer-dependence as an interpretive starting point or, in 't Hooft's case, derive it from a specific microphysical model. The present framework differs in that it identifies *necessary and sufficient conditions* under which any embedded observer in any deterministic system necessarily sees quantum mechanics, independent of the system's specific physical content—and proves that these are the *only* conditions under which quantum mechanics arises from a deterministic embedding.

The structural constraints imposed by this reduction are not merely interpretive; they carry massive empirical weight. By establishing this equivalence, the framework reveals that phenomena previously viewed as catastrophic theoretical anomalies—most notably the 10^{122} cosmological constant discrepancy—are in fact mandatory, predictable consequences of embedded observation.

1.2 Axioms

The framework rests on three axioms. None reference quantum mechanics, general relativity, or any specific physical theory.

1. **Deterministic dynamics.** The total system evolves deterministically on a phase space Γ. The phase-space density $\rho(q, p, t)$ evolves via the Liouville equation:

$$\frac{\partial \rho}{\partial t} = \{H_{\text{tot}}, \rho\} \equiv \mathcal{L}\rho \tag{1}$$

   where $\mathcal{L}$ is the Liouville operator and $\{\cdot, \cdot\}$ denotes the Poisson bracket.

Finiteness condition. The effective configuration space of the visible sector is finite-dimensional. A discreteness scale ϵ provides a finite minimal cell volume, ensuring that any finite spatial region contains at most a finite number of independent cells. In Part I, ϵ is a free parameter; its value is determined *a posteriori* by self-consistency (§6), which pins $\epsilon = 2\,l_p$ exactly. In Part II, the finiteness condition is not an additional assumption but a consequence of the partition geometry: the cosmological horizon has finite area $A = 4\pi c^2/H^2$ (from GR alone), and the number of modes that couple through a finite-area boundary is A/ϵ^2, which is finite for any $\epsilon > 0$. The stochastic process that the trace-out generates therefore lives on a finite-dimensional space regardless of whether the total phase space extends infinitely beyond the horizon. This condition enters the proof at a single point: the stochastic-quantum correspondence (§3.1), which requires a finite configuration space. Independent support comes from holographic arguments that the Hilbert space of quantum gravity is locally finite-dimensional [27], establishing that this condition aligns naturally with known holographic bounds rather than serving as an ad hoc lattice assumption.

2. **Causal partition.** Γ is partitioned into a visible sector Γ_V accessible to the observer and a hidden sector Γ_H that is not:

$$\Gamma = \Gamma_V \times \Gamma_H \tag{2}$$

The partition is fixed on timescales relevant to the observer (it may drift on much longer timescales, but such drift is not required for the theorem). The total Hamiltonian decomposes as:

$$H_{\text{tot}} = H_V + H_H + H_{\text{int}} \tag{3}$$

The product decomposition $\Gamma_V \times \Gamma_H$ is a simplification. A real observer's epistemic access is graded: some degrees of freedom are measured precisely, others poorly, and others not at all. The axiom requires that this graded access be well-approximated by a sharp partition—that there exist a clean division into accessible and inaccessible sectors such that cross-partition correlations enter only through H_{int}. This is a condition on the physics, not just the observer: it is satisfied when a physical boundary (causal, thermodynamic, or spatial) enforces a separation sharp enough that the residual leakage across the boundary can be absorbed into the interaction term. The theorem applies to any system where such an approximation holds; Part II identifies a physical realization where the sharpness is exact.

3. **Classical probability.** The observer's predictions are classical expectation values over the phase space; standard results of probability theory (the law of large numbers, the central limit theorem) are the primary statistical tools.

These axioms contain no quantum postulates—no Hilbert spaces, no commutation relations, no Born rule. The framework's central claim is that quantum mechanics *emerges* from these premises under the conditions specified below (§1.3).

1.3 Conditions on the Hidden Sector

The theorem requires three conditions on the relationship between the visible and hidden sectors. These are stated as sufficient conditions for the emergence of quantum mechanics; §3.3 proves they are also necessary, establishing a full equivalence.

(C1) Non-zero coupling. The interaction Hamiltonian satisfies $H_{\text{int}} \neq 0$. The visible and hidden sectors are dynamically coupled, not merely co-existing. The coupling is bidirectional: visible-sector transitions influence the hidden-sector state, and the hidden-sector state influences visible-sector transition probabilities.

(C2) Slow-bath timescale separation. The hidden sector's correlation time τ_B vastly exceeds the visible sector's dynamical timescale τ_S:

$$\tau_S \ll \tau_B \tag{4}$$

This is the *inverse* of the standard Markovian regime, which requires $\tau_B \ll \tau_S$. The hidden sector evolves on timescales far exceeding those of any process accessible to the observer.

(C3) Sufficient capacity. The hidden sector has enough degrees of freedom that visible-sector interactions do not appreciably perturb its state on timescales $\ll \tau_B$. Precisely: if N_H and N_V denote the number of independent degrees of freedom in the hidden and visible sectors respectively, then N_H/N_V is large enough that the cumulative imprint of all visible-sector transitions during any observationally relevant interval occupies a negligible fraction of the hidden sector's state space.

Statement of the theorem. Under Axioms 1–3, the finiteness condition, and conditions (C1)–(C3), the embedded observer's reduced description of the visible sector is mathematically equivalent to unitarily evolving quantum mechanics. The remainder of Part I constitutes the proof. The conditions are shown to be not merely sufficient but *necessary* for P-indivisibility in any deterministic dilation (§3.3), establishing a full equivalence between quantum mechanics and embedded observation.

1.4 Partition-Relativity

Before proceeding to the proof, a structural consequence of the axioms is derived that will constrain the emergent description throughout.

Lemma 1.1 (Partition-relativity)**.** *The emergent description of the visible sector is uniquely determined by the partition. Two observers whose causal access defines the same partition*

obtain the same reduced description. Any parameters of the emergent theory can depend only on quantities intrinsic to the partition—the geometric and thermodynamic properties of the boundary between visible and hidden sectors.

Proof. By Axiom 1, the dynamics are deterministic: the Hamiltonian flow ϕ_t on $\Gamma_V \times \Gamma_H$ is a single, uniquely defined map. By Axiom 2, the partition $\Gamma = \Gamma_V \times \Gamma_H$ is fixed. By Axiom 3, the observer uses the Liouville measure μ on Γ_H—the unique absolutely continuous measure preserved by the Hamiltonian flow (Liouville's theorem); any other smooth measure would evolve under the dynamics, introducing dependence on an arbitrary reference time, and singular invariant measures (e.g., those supported on periodic orbits) are excluded by Axiom 3's requirement that the observer assigns classical probability distributions over the full phase space. The marginalized transition probabilities

$$T_{ij}(t_2, t_1) = \int_{\Gamma_H} \delta_{x_j}[\pi_V(\phi_{t_2-t_1}(x_i, h))]\, d\mu(h) \tag{5}$$

are therefore uniquely determined by three inputs: the dynamics (ϕ_t), the partition (Γ_V, Γ_H), and the measure (μ). Since ϕ_t and μ are fixed by the axioms, the transition probabilities—and hence the complete stochastic process—depend only on the partition. If this process is P-indivisible (as shown in §2), the stochastic-quantum correspondence maps it to a quantum description. Since the stochastic process is unique, the quantum description is unique (up to unitary equivalence and gauge freedoms resolved below). In particular, any free parameters of the emergent theory—including the action scale $\hbar$—can depend only on quantities intrinsic to the partition: the geometric and thermodynamic properties of the boundary between Γ_V and Γ_H. □

Why this matters. Partition-relativity is not an additional postulate but a direct consequence of the axioms' deterministic and measure-theoretic content. It will play three roles in what follows: (i) it ensures that the emergent quantum mechanics is universal for all sub-c observers sharing the same cosmological horizon (§4.1); (ii) it constrains the action scale $\hbar$ to depend only on quantities intrinsic to the partition, and specifically to its local *structure* rather than its volume, which together with dimensional analysis and thermal self-consistency determines the exact value $\hbar = c^3\epsilon^2/(4G)$ (§5.2); and (iii) it provides the logical bridge between the general theorem and the quantitative derivation: the same principle that guarantees uniqueness also restricts which physical quantities can appear in the emergent description.

2. Emergent Stochasticity and P-Indivisibility

2.1 Emergent Stochasticity

The total system (visible + hidden sectors) evolves deterministically (Axiom 1). Every phase-space point $(x, h) \in \Gamma_V \times \Gamma_H$ maps to exactly one future point under the Hamiltonian flow. No randomness exists at this level. But the visible sector alone—what the embedded observer accesses—is stochastic. The observer knows x but not h. Different hidden-sector states h_k compatible with the same x send x to different visible-sector futures x'_k. The observer must assign transition probabilities $T_{ij}(t_2, t_1)$ by marginalizing the deterministic flow over the hidden sector using the Liouville measure, as defined in §1.4. As established there, the Liouville measure is the unique absolutely continuous measure preserved by the Hamiltonian flow, ensuring that the transition probabilities—and hence the emergent description—are uniquely determined by the partition.

This architecture makes the framework a hidden variable theory: the randomness observed by the embedded observer is epistemic, arising from ignorance of hidden-sector degrees of freedom rather than fundamental indeterminacy. The consistency of this structure with Bell's theorem is addressed in §3.2.

2.2 The Slow-Bath Regime

The Markovian limit for reduced dynamics requires a strict timescale separation $\tau_B \ll \tau_S$ [4], where the environment decorrelates much faster than the system evolves. Condition (C2) inverts this hierarchy: $\tau_S \ll \tau_B$. The hidden sector evolves on timescales far exceeding those of any visible-sector process, so its correlations do not decorrelate between successive system transitions. The memory kernel of the reduced dynamics is effectively constant over all observationally relevant timescales.

A slow bath must be distinguished from a static external field. A genuinely static (decoupled) hidden sector would merely shift Hamiltonian parameters in H_V without inducing information backflow. By condition (C1), the hidden sector is not decoupled: the interaction Hamiltonian H_{int} is continuously active. During each visible-sector transition, H_{int} imprints the identity of the transition onto the hidden-sector state. Because the hidden sector is slow—not static—these imprints persist without being thermally overwritten: the hidden sector's internal relaxation timescale (τ_B) vastly exceeds the interval between successive visible-sector transitions (τ_S). On subsequent transitions, the coupling reads back the stored correlations, producing transition probabilities that depend on the system's prior trajectory—the non-Markovian open-systems regime of Breuer and Petruccione [4].

2.3 P-Indivisibility

Theorem 2.1 (P-indivisibility of embedded observation)**.** *Let $\mathcal{C}_V$ and $\mathcal{C}_H$ be finite sets with $|\mathcal{C}_V| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection. Define the one-step transition matrix on $\mathcal{C}_V$ by marginalizing over $\mathcal{C}_H$ with respect to the uniform measure:*

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi(x_i, h)) = x_j\}|}{|\mathcal{C}_H|} \tag{6}$$

and the k-step transition matrix $T_{ij}^{(k)}$ by the same construction applied to φ^k. If T is not a permutation matrix—i.e., the marginalized dynamics is genuinely stochastic—then the stochastic process generated by iterating φ is P-indivisible.

Proof. The proof uses the total variation distance $d(p, q) = \frac{1}{2}\sum_k |p_k - q_k|$ between probability distributions. A stochastic process is P-divisible if and only if $d(p(t), q(t))$ is non-increasing in t for all initial distributions p, q [6, 7]. P-indivisibility is established by exhibiting a non-monotonicity.

Step 1 (Recurrence). Since φ is a bijection on a finite set, it has finite order: there exists $N \geq 1$ such that $\varphi^N = \mathrm{id}$. Every state (x_i, h) returns to itself after N steps, so $T^{(N)} = I$. In particular, $d(\delta_i T^{(N)}, \delta_j T^{(N)}) = d(\delta_i, \delta_j) = 1$ for all $i \neq j$.

Step 2 (Strict contraction at step 1). Since T is row-stochastic and not a permutation matrix, there exist distinct indices i, j whose rows have overlapping support: some column l satisfies $T_{il} > 0$ and $T_{jl} > 0$. For such i, j:

$$d(\delta_i T, \delta_j T) = \frac{1}{2}\sum_k |T_{ik} - T_{jk}| < \frac{1}{2}\sum_k (T_{ik} + T_{jk}) = 1 \tag{7}$$

The strict inequality holds because every column l with overlapping support contributes $|T_{il} - T_{jl}| < T_{il} + T_{jl}$ to the sum.

Step 3 (Non-monotonicity). Steps 1 and 2 together give:

$$d(\delta_i T^{(1)}, \delta_j T^{(1)}) < 1 = d(\delta_i T^{(N)}, \delta_j T^{(N)}) \tag{8}$$

The total variation distance decreases from step 0 to step 1, then increases from step 1 to step N. This is non-monotonic: the process is P-indivisible. □

The theorem requires only two inputs: deterministic bijective dynamics (Axiom 1) and non-trivial coupling (condition C1, which ensures T is not a permutation matrix). The finiteness condition guarantees the finite configuration space on which the theorem operates. The proof mechanism is information backflow: information about the initial visible-sector state leaks to the hidden sector via the coupling (Step 2) and is later read back by the deterministic dynamics (Step 1), producing the non-monotonic trace distance that characterizes P-indivisibility.

Continuous-time extension. The theorem is stated for discrete-time dynamics, but the physical setup involves continuous-time Hamiltonian flow. The extension follows from continuity and Poincaré recurrence. The Hamiltonian flow φ_t on the finite-dimensional phase space (finiteness condition) preserves the Liouville measure on compact energy surfaces. The transition matrix $T_{ij}(t)$ is continuous in t, with $T(0) = I$. Non-zero coupling (C1) ensures that $T(t)$ departs from the permutation class for $t > 0$, so that $d(\delta_i T(t), \delta_j T(t)) < 1$ for some pair i, j and some $t > 0$. By Poincaré's recurrence theorem, for any $\delta > 0$, there exists $t_R > 0$ such that the set of hidden-sector states satisfying $\pi_V(\varphi_{t_R}(x_i, h)) = x_i$ has measure $> 1 - \delta$, giving $d(\delta_i T(t_R), \delta_j T(t_R)) > 1 - 2\delta$. For δ small enough, this exceeds $d(\delta_i T(t), \delta_j T(t))$, establishing non-monotonic trace distance and hence P-indivisibility for the continuous-time process.

Convergence with prior work. The theorem converges with several independent lines of work: Pechukas [5] showed that reduced dynamics of a correlated system need not preserve positivity; Rivas, Huelga, and Plenio [6, 7] formalized the connection between divisibility failure and information backflow; Pollock et al. [8] derived necessary and sufficient Markovianity conditions via the process tensor framework; and most directly, Strasberg and Esposito [9] demonstrated non-Markovian reduced dynamics precisely in the slow-bath regime of condition (C2).

The physical role of conditions (C2) and (C3). The theorem establishes P-indivisibility from (C1) and finiteness alone. Conditions (C2) and (C3) are not needed for the mathematical result but are essential for its physical force. (C2) ensures that information backflow occurs on timescales accessible to the observer, not merely at Poincaré recurrence times that may exceed the age of the universe. (C3) ensures that P-indivisibility persists at physical scale: if the hidden sector has insufficient capacity, successive imprints overwrite earlier ones—the hidden sector saturates, and the reduced dynamics cross over to effectively Markovian behavior. The conjunction of (C2) and (C3) guarantees that P-indivisibility is strong, persistent, and observationally relevant.

2.4 Explicit Construction: A Minimal Model

The following minimal model illustrates the theorem and makes its mechanism explicit.

The model. The visible sector has two states: $x \in \{0, 1\}$. The hidden sector has six states: $h \in \{1, 2, 3, 4, 5, 6\}$. The total phase space has 12 states (x, h). The dynamics are a deterministic, invertible permutation σ on these 12 states, defined as six disjoint transpositions:

$$\sigma = (0{,}1 \leftrightarrow 1{,}1)\ (0{,}2 \leftrightarrow 1{,}2)\ (0{,}3 \leftrightarrow 0{,}4)\ (0{,}5 \leftrightarrow 0{,}6)\ (1{,}3 \leftrightarrow 1{,}4)\ (1{,}5 \leftrightarrow 1{,}6) \tag{9}$$

The first two transpositions couple the visible and hidden sectors—they change x while preserving h, implementing condition (C1). The remaining four permute hidden-

sector labels within each visible-sector value, representing the hidden sector's internal dynamics. Since σ is a product of disjoint transpositions, $\sigma^2 = \mathrm{id}$: the hidden sector retains perfect correlations between successive transitions, satisfying condition (C2). Six hidden-sector states per visible-sector value ensures the hidden sector is not saturated by a single transition, satisfying condition (C3).

Transition matrices. At $t = 0$, the observer knows x but not h, and assigns the uniform (Liouville) measure over the 6 hidden states. At $t = 1$ (one application of σ):

From $x = 0$: two of six hidden states ($h = 1, 2$) send x to 1; four ($h = 3, 4, 5, 6$) leave x at 0. From $x = 1$: two of six ($h = 1, 2$) send x to 0; four leave x at 1. The one-step transition matrix is:

$$T(1,0) = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix} \tag{10}$$

At $t = 2$ ($\sigma^2 = \mathrm{id}$): every state returns to itself, so $T(2,0) = I$.

A Markov chain with transition matrix $T(1,0)$ would predict $T(2,0)_{\text{Markov}} = T(1,0)^2 = \left(\begin{smallmatrix} 5/9 & 4/9 \\ 4/9 & 5/9 \end{smallmatrix}\right)$—continued mixing. The actual dynamics show complete *un-mixing*: the system returns to its initial state with certainty. This is information backflow.

P-divisibility test. If the process were P-divisible, one could write $T(2,0) = \Lambda(2,1) \cdot T(1,0)$ with $\Lambda(2,1)$ a valid stochastic matrix (non-negative entries, rows summing to 1). Computing:

$$\Lambda(2,1) = T(2,0) \cdot [T(1,0)]^{-1} = I \cdot \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix} \tag{11}$$

The matrix $\Lambda(2,1)$ has negative entries (-1). No valid stochastic matrix can serve as the intermediate propagator. **The process is P-indivisible.** □

The mechanism. The hidden sector acts as a memory register. At step 1, the transpositions $(0,h) \leftrightarrow (1,h)$ for $h \in \{1,2\}$ flip x while preserving h. After step 1, a state now at $x = 1$ with $h \in \{1,2\}$ carries the record "I was at $x = 0$." At step 2, the same transposition reads this record and flips x back. The visible-sector dynamics exhibit information backflow: the system appears to "remember" its origin, which is impossible for a memoryless process. The model illustrates the mechanism in the idealized limit of perfect recurrence ($\sigma^2 = \mathrm{id}$, i.e., $\tau_B/\tau_S = 2$). The cosmological realization of Part II has $\tau_S/\tau_B \sim 10^{-32}$; the P-indivisibility is more extreme, not less, when the hidden sector's memory is more persistent.

3. The Emergence of Quantum Mechanics

3.1 The Stochastic-Quantum Correspondence

By Barandes' stochastic-quantum correspondence [10, 11], any indivisible stochastic process on a finite configuration space of size n can be embedded into a unitarily evolving quantum system on a Hilbert space of dimension $\leq n^3$. A stochastic process on n configurations is characterized by its transition matrices $T_{ij}(t_2, t_1)$. If the process is P-divisible, these matrices can be factored through intermediate times and the system admits a classical description at all scales. If it is P-indivisible, no such factoring exists: the transition matrices at fine time resolution develop negative entries (as in §2.4), violating classicality. Barandes shows that the Stinespring dilation theorem then guarantees the existence of a Hilbert space and a unitary operator $U(t)$ such that $T_{ij}(t) = |U_{ij}(t)|^2$—the transition probabilities of the stochastic process are exactly the Born-rule probabilities of the quantum system. Indivisibility is not merely *analogous* to quantum behavior; it is *mathematically equivalent* to it. The P-indivisible process established in §2.3 satisfies the correspondence's hypotheses directly.

From phase space to configuration space. The axioms define the dynamics on a phase space Γ with coordinates (q, p), while Barandes' correspondence operates on a configuration space. The connection is the marginalization itself: the transition probabilities $T_{ij}(t_2, t_1)$ defined in §2.1 are projections of the full phase-space flow onto the configuration sector of Γ_V, with the momenta—along with all hidden-sector degrees of freedom—absorbed into the marginalization over the Liouville measure. The resulting stochastic process lives on the discrete configuration space defined by the finiteness condition, where the correspondence applies directly.

Three features of quantum mechanics emerge from this correspondence. The Schrödinger equation is the unique time-evolution law, arising from the differentiability of the unitary in time. The Born rule is not an additional postulate but the equilibrium distribution of the indivisible stochastic process [10, 11]. The action scale $\hbar$ requires more careful treatment. The correspondence guarantees the existence of a unitary evolution operator $U(t)$, but U is dimensionless; $\hbar$ enters only when one defines the Hamiltonian $\hat{H}(t) \equiv i\hbar\,\partial_t U \cdot U^\dagger$, converting the dimensionless rate of change into energy units. The correspondence itself is silent about the value of this conversion factor—it establishes that a quantum description exists for any P-indivisible process, but not which action scale it carries. What fixes $\hbar$ is the physical content of the partition: the emergent description must be unique for a given partition (§1.4), so the action scale can depend only on quantities intrinsic to the partition boundary. Its value is derived for the cosmological realization in Part II (§5).

Finite-dimensionality. Barandes' proof is established for finite-dimensional configuration spaces. In the general theorem (Part I), the finiteness condition guarantees

applicability directly: the finite minimal cell volume ensures that any finite spatial region contains at most a finite number of independent cells, and the causal partition (Axiom 2) restricts the visible sector to a finite total volume. In the cosmological application (Part II), the same guarantee comes from the partition geometry: the horizon's finite area bounds the number of coupled modes to A/ϵ^2, ensuring the effective configuration space is finite-dimensional without requiring the total phase space to be finite. Calvo [12] has extended the correspondence to infinite-dimensional systems, relaxing the finite-dimensionality requirement, but the extension is not required here.

3.2 Bell Inequality Violations

The framework is a hidden variable theory: quantum randomness arises epistemically from ignorance of hidden-sector degrees of freedom. Bell's theorem [13] rules out *local* hidden variable theories satisfying the factorizability condition $P(a, b \mid x, y, \lambda) = P(a \mid x, \lambda) \cdot P(b \mid y, \lambda)$. The present framework evades Bell's theorem not by violating measurement independence (superdeterminism) but by violating factorizability through P-indivisible joint dynamics—while remaining causally local.

The mechanism is non-factorizable joint dynamics from a shared causal past. Two subsystems Q and R that interact at preparation and are subsequently separated carry a joint transition matrix $T_{QR}(t_2, t_1)$ on $\mathcal{C}_Q \times \mathcal{C}_R$ that does not factorize:

$$T_{QR} \neq T_Q \otimes T_R \tag{12}$$

This non-factorizability *is* entanglement at the stochastic level [10, 11, 14]. The correlations are encoded in the structure of the joint transition matrix rather than in a hidden variable λ over which one could integrate. Since the process is indivisible, there are no well-defined intermediate conditional probabilities that would permit the factorization Bell's theorem requires.

In the Jarrett decomposition, quantum mechanics violates outcome independence (OI) while preserving parameter independence (PI). The present framework reproduces this structure exactly. Barandes, Hasan, and Kagan [14] prove that the conjunction of indivisibility and causal locality restricts the maximum CHSH correlator to exactly Tsirelson's bound $|S| \leq 2\sqrt{2}$, with independent support from Le et al. [15]. Measurement independence is preserved by construction: transition probabilities are defined independently of the initial distribution. This placement—violating outcome independence while preserving measurement independence and parameter independence—is precisely the class Fine's theorem [24] permits.

3.3 The Characterization Theorem: Necessity of the Conditions

The theorem of §1.3 establishes sufficiency: the axioms, finiteness, and conditions (C1)–(C3) guarantee quantum mechanics. This section proves the converse—that quantum mechanics *requires* these conditions—establishing a full equivalence between quantum mechanics and embedded observation.

The logical structure. Barandes' stochastic-quantum correspondence is already bidirectional: a stochastic process is P-indivisible if and only if it is mathematically equivalent to unitarily evolving quantum mechanics [10, 11]. The equivalence

$$\text{QM} \iff \text{P-indivisibility} \tag{13}$$

is established. What remains is to prove the converse of the other link:

$$\text{P-indivisibility} \implies \text{embedded observation under (C1)–(C3)} \tag{14}$$

The full equivalence QM $\iff$ embedded observation (under the axioms, finiteness, and (C1)–(C3)) then follows by transitivity.

Dilation existence (necessity of the axioms). Any stochastic process on a finite configuration space can be realized as the marginalization of a deterministic process on a larger state space: one enlarges the state space to encode the randomness as ignorance of auxiliary variables and recovers the original process by tracing them out. This is a standard construction in classical probability theory. The dilation furnishes a deterministic total system (Axiom 1), a partition into visible and hidden sectors (Axiom 2), and classical marginalization (Axiom 3). The three axioms are therefore necessary in the sense that any stochastic process—and hence any P-indivisible process—admits a deterministic embedding with a causal partition. The finiteness of the configuration space is inherited from the original process.

Theorem 3.1 (C1 equivalence)**.** *Let $\mathcal{C}_V$ and $\mathcal{C}_H$ be finite sets with $|\mathcal{C}_V| \geq 2$, and let $\varphi : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection. Define the stochastic process on $\mathcal{C}_V$ by marginalizing over $\mathcal{C}_H$ with respect to the uniform measure. The process is P-indivisible if and only if* (C1) *holds—that is, if and only if the marginalized one-step transition matrix T is not a permutation matrix.*

Proof. The forward direction (C1 $\implies$ P-indivisibility) is the theorem of §2.3. The reverse is proved by contraposition. If T is a permutation matrix, then T represents a deterministic map $\sigma : \mathcal{C}_V \to \mathcal{C}_V$ with $T_{ij} = \delta_{j,\sigma(i)}$. The k-step transition matrix is $T^{(k)}_{ij} = \delta_{j,\sigma^k(i)}$, also a permutation matrix for every k. The intermediate propagator $\Lambda(k_2, k_1) = T^{(k_2)} \cdot [T^{(k_1)}]^{-1} = T^{(k_2-k_1)}$ is a permutation matrix and hence a valid stochastic matrix (non-negative entries, rows summing to 1) for all $k_2 > k_1$. The process is P-

divisible. Contrapositively, P-indivisibility requires T not to be a permutation matrix, which requires non-trivial coupling: $H_{\text{int}} \neq 0$. □

Theorem 3.2 (C2 necessity: Markovian limit)**.** *Let $\varphi_{\text{coup}} : \mathcal{C}_V \times \mathcal{C}_H \to \mathcal{C}_V \times \mathcal{C}_H$ be a bijection implementing the visible-hidden coupling, and let $\varphi_H : \mathcal{C}_H \to \mathcal{C}_H$ be an ergodic permutation (a single cycle of length $|\mathcal{C}_H|$). Define the interleaved dynamics $\Phi_m = (\text{id}_V \times \varphi_H^m) \circ \varphi_{\text{coup}}$, where m represents the number of hidden-sector mixing steps between coupling events (m is the ratio τ_S/τ_B, so large m represents the fast-bath regime $\tau_B \ll \tau_S$). Let $T_m^{(k)}$ denote the k-step visible-sector transition matrix of Φ_m. Then for m coprime to $|\mathcal{C}_H|$, the process $\{T_m^{(k)}\}$ converges to a Markov chain as $|\mathcal{C}_H| \to \infty$ with the coupling structure held fixed. In the Markovian limit—when the hidden sector's state is independent of the visible sector's history before each coupling event—the reduced process is exactly P-divisible.*

Proof. The proof proceeds in two parts: first, that the idealized Markovian limit is P-divisible; second, that fast mixing drives the system to this limit.

Part 1 (Markovian limit is P-divisible). Consider the idealized process in which the hidden-sector state is drawn independently from the uniform measure on $\mathcal{C}_H$ before each coupling event, regardless of the visible sector's history. The one-step transition matrix is:

$$T_{ij} = \frac{|\{h \in \mathcal{C}_H : \pi_V(\varphi_{\text{coup}}(x_i, h)) = x_j\}|}{|\mathcal{C}_H|} \tag{15}$$

Since the hidden sector is re-randomized at each step, the transition probability at step k depends only on the current visible state, not on the history. The k-step transition matrix is therefore $T^{(k)} = T^k$ (the k-th matrix power). The intermediate propagator is $\Lambda(k_2, k_1) = T^{k_2 - k_1}$, which is a stochastic matrix for all $k_2 > k_1$ (any power of a stochastic matrix is stochastic). The process is P-divisible.

Part 2 (Fast mixing approaches the Markovian limit). Since φ_H is an ergodic permutation (a single cycle of length $|\mathcal{C}_H|$), the map φ_H^m with m coprime to $|\mathcal{C}_H|$ is also a single cycle of length $|\mathcal{C}_H|$. For any initial distribution μ on $\mathcal{C}_H$, repeated application of an ergodic permutation drives μ toward the uniform distribution in the Cesàro sense: the time-averaged distribution $\bar{\mu}_N = \frac{1}{N}\sum_{k=0}^{N-1}(\varphi_H^m)_*^k \mu$ converges to the uniform distribution μ_{unif} as $N \to \infty$.

In the interleaved dynamics, between the $(k-1)$-th and k-th coupling events, the hidden sector undergoes m steps of φ_H. If m is large relative to $|\mathcal{C}_H|$, the hidden-sector distribution—conditioned on the visible-sector state—approaches uniformity before the next coupling event. Formally: for any $\delta > 0$ and any initial distribution μ_0 on $\mathcal{C}_H$, the total variation distance satisfies $d((\varphi_H^m)_*\mu_0, \mu_{\text{unif}}) < \delta$ whenever m contains a full cycle (i.e., $m \geq |\mathcal{C}_H|$, by the ergodic theorem for finite groups). In the limit where the mixing is complete, the hidden-sector distribution before each coupling is exactly uniform and independent of the visible-sector history. The transition probabilities at each step are then exactly T_{ij}, independent of the past, and the process reduces to Part 1.

Contrapositive (necessity of C2). If the process is P-indivisible, Part 1 shows that the hidden sector's state cannot be independent of the visible sector's history at the moment of each coupling. The hidden sector must retain memory of past visible-sector transitions—correlations that have not been overwritten by internal mixing. Part 2 shows that fast mixing ($\tau_B \ll \tau_S$, large m) drives these correlations to zero. Therefore P-indivisibility requires that the hidden sector's mixing timescale exceed the coupling timescale: $\tau_S \ll \tau_B$, which is condition (C2). □

Remark. The theorem formalizes the physical distinction between a slow bath and a fast bath. A slow bath (C2 satisfied) retains the imprints of past visible-sector transitions and reads them back on subsequent steps—the mechanism underlying information backflow. A fast bath (C2 violated) erases those imprints through internal mixing before they can be read back, destroying the memory that P-indivisibility requires. The inverted timescale hierarchy is not an incidental feature of the sufficiency proof; it is the essential condition that enables the hidden sector to function as a persistent memory register.

Theorem 3.3 (C3 necessity: information-theoretic capacity bound)**.** *Let φ be a bijection on $\mathcal{C}_V \times \mathcal{C}_H$ with $|\mathcal{C}_V| = n$ and $|\mathcal{C}_H| = m$. Let $X_0, X_1, X_2, \ldots$ denote the visible-sector trajectory and $H_0, H_1, H_2, \ldots$ the corresponding hidden-sector trajectory under the deterministic dynamics. The non-Markovian mutual information—the information that the visible sector's history provides about its future evolution, beyond what is contained in its present state—satisfies:*

$$I(X_{<t}; X_{>t} \mid X_t) \leq \log_2 m \tag{16}$$

In particular, the total non-Markovian information capacity of the reduced process is bounded by the logarithmic size of the hidden sector.

Proof. The total system evolves deterministically: X_{t+1} and H_{t+1} are deterministic functions of (X_t, H_t). Therefore the entire future trajectory $X_{>t} = (X_{t+1}, X_{t+2}, \ldots)$ is a deterministic function of (X_t, H_t). By the data processing inequality for Markov chains: if $A \to B \to C$ is a Markov chain, then $I(A; C) \leq I(A; B)$. Conditioning on X_t:

$$X_{<t} \to H_t \to X_{>t} \quad \text{(conditioned on } X_t\text{)} \tag{17}$$

This is a Markov chain because, given X_t and H_t, the future $X_{>t}$ is fully determined and independent of $X_{<t}$. Applying the data processing inequality:

$$I(X_{<t}; X_{>t} \mid X_t) \leq I(X_{<t}; H_t \mid X_t) \leq H(H_t \mid X_t) \leq \log_2 |\mathcal{C}_H| = \log_2 m \tag{18}$$

where the second inequality is a standard property of mutual information ($I(A; B) \leq H(B)$), and the third is the maximum entropy of a random variable on m values. □

Corollary 3.4 (Necessity of C3)**.** *P-indivisibility across all n visible-sector configurations requires non-Markovian correlations spanning the full configuration space. The non-Markovian information required to sustain P-indivisibility across n modes scales as $\Omega(\log n)$: distinguishing the non-Markovian evolution of n visible-sector states requires at least $\log_2 n$ bits of hidden-sector memory. By the capacity bound, this requires $\log_2 m \geq \Omega(\log n)$, i.e., $m \geq n^\alpha$ for some $\alpha > 0$. The hidden sector must scale with the visible sector—precisely condition (C3).*

Proof. Suppose the visible sector has n configurations and the reduced process is P-indivisible with information backflow affecting all n configurations. The non-monotonicity of the trace distance (§2.3) requires that the hidden sector stores which of the n visible-sector states the system previously occupied—or more generally, stores a trajectory-dependent record that affects the future transition probabilities of each visible-sector state differently. Storing a record that distinguishes n possible states requires at least $\log_2 n$ bits. By the capacity bound theorem, the total non-Markovian information is at most $\log_2 m$. Therefore:

$$\log_2 m \geq \log_2 n \implies m \geq n \tag{19}$$

If the P-indivisibility is to persist over K successive transitions, the bound tightens to $\log_2 m \geq K \cdot f(n)$ for some function f depending on the strength of the correlations, requiring m/n to grow with the observation window. In particular, if the observation window includes $K \sim n$ transitions, the hidden sector cannot absorb these imprints without saturation unless m/n is large. This is condition (C3).

Contrapositive: if m is not sufficiently large relative to n, the $\log_2 m$ capacity bound chokes the non-Markovian information flow before it can sustain P-indivisibility across the full visible sector. The hidden sector saturates—successive imprints overwrite earlier ones—and the reduced dynamics on the affected configurations cross over to effective P-divisibility. □

Remark. The capacity bound is *information-theoretic*—it does not depend on the details of the bijection φ, only on the cardinalities of the two sectors. This universality is what makes C3 a necessary condition rather than a model-dependent feature: *any* deterministic dilation of a P-indivisible process on n configurations must have a hidden sector of at least n states, and a hidden sector much larger than n for the P-indivisibility to persist under repeated observation. The bound is tight in the sense that the minimal model of §2.4 achieves P-indivisibility with $m = 6$ for $n = 2$, consistent with $m \geq n$.

Characterization theorem (equivalence of quantum mechanics and embedded observation). Let $\mathcal{C}_V$ be a finite configuration space with $|\mathcal{C}_V| \geq 2$, and let $\mathcal{S}$ be a stochastic process on $\mathcal{C}_V$. The following are equivalent:

(i) $\mathcal{S}$ is mathematically equivalent to unitarily evolving quantum mechanics (via Barandes' stochastic-quantum correspondence [10, 11]).

(ii) $\mathcal{S}$ is P-indivisible.

(iii) $\mathcal{S}$ arises from the marginalization of a deterministic bijection φ on $\mathcal{C}_V \times \mathcal{C}_H$ over a hidden sector $\mathcal{C}_H$, with:

- (C1) non-trivial coupling: the marginalized transition matrix T is not a permutation matrix;
- (C2) slow-bath memory: the hidden sector retains correlations between successive coupling events;
- (C3) sufficient capacity: $|\mathcal{C}_H| \geq |\mathcal{C}_V|$, with $|\mathcal{C}_H|/|\mathcal{C}_V|$ large enough that the hidden sector is not saturated.

Proof. (i) $\iff$ (ii) is Barandes' stochastic-quantum correspondence [10, 11]. (iii) $\implies$ (ii) is the theorem of §2.3. (ii) $\implies$ (iii) is proved by the three necessity theorems above: dilation existence provides the deterministic embedding and the axioms; the C1 equivalence theorem establishes that the coupling must be non-trivial; the C2 necessity theorem establishes that the hidden sector must retain memory; and the C3 capacity bound establishes that the hidden sector must have sufficient dimensionality. □

Interpretive consequence. The characterization theorem closes the logical gap that a sufficiency-only result would leave open. Quantum mechanics is not merely *compatible with* the epistemic situation of an embedded observer—it is *equivalent to* it. Every quantum system can be understood as one in which an observer is embedded in a larger deterministic system with non-trivial, slowly decorrelating coupling to a hidden sector of sufficient capacity. Conversely, every such embedding necessarily produces quantum mechanics in the observer's reduced description.

PART II: THE COSMOLOGICAL APPLICATION

4. The Cosmological Horizon as Causal Partition

4.1 The Partition

General relativity defines the causal structure of spacetime. The cosmological horizon is the boundary beyond which no signal propagating at or below c can reach the em-

bedded observer. Every physically realizable measurement technique—electromagnetic, gravitational-wave, neutrino—propagates at or below c and therefore encounters the same boundary. The horizon implements the causal partition of Axiom 2 for all sub-c observers simultaneously: the visible sector Γ_V is the interior of the cosmological horizon, and the hidden sector Γ_H is everything beyond it.

This identification is not a modeling choice but a consequence of GR's causal structure. A hypothetical observer with access to a causal channel not constrained by the light cone would have a different partition, a different (or absent) hidden sector, and by the theorem's own logic, a different emergent description. No such channel is available within GR.

4.2 Verification of the Conditions

Condition (C1): Non-zero coupling. The cosmological horizon is not a static wall. Stress-energy conservation across the horizon enforces dynamical correlations between the visible and hidden sectors: matter and radiation cross the horizon continuously, and the horizon area responds to changes in the interior energy density via the Friedmann equation. The coupling is bidirectional. **Condition (C1) is satisfied.**

Condition (C2): Slow-bath timescale separation. The hidden sector's shortest correlation timescale is set by the Hubble time: $\tau_B \gtrsim 1/H \sim 10^{17}$ s, since modes beyond the horizon evolve on timescales no shorter than the rate at which the horizon itself changes. For any laboratory-scale process with characteristic frequency $\omega \gg H$, the visible sector's dynamical timescale is $\tau_S \sim 1/\omega \sim 10^{-15}$ s or shorter. The ratio is:

$$\frac{\tau_S}{\tau_B} \sim 10^{-32} \tag{20}$$

The hidden sector cannot decorrelate between any pair of visible-sector events. The Markovian approximation fails categorically. **Condition (C2) is satisfied.**

Condition (C3): Sufficient capacity. The hidden sector has $N_{\text{modes}} = A/\epsilon^2$ independent degrees of freedom, where $A = 4\pi c^2/H^2$ is the horizon area and ϵ is the discreteness scale of the finiteness condition. For any ϵ at or above the Planck scale, the horizon area is cosmological ($A \sim 10^{52}$ m^2) and the visible sector involves at most $\sim 10^{80}$ baryons. No laboratory process can appreciably perturb the hidden sector's state. The hidden sector is never saturated. **Condition (C3) is satisfied.**

4.3 Application of the Theorem

All three axioms hold. All three conditions hold. The finiteness condition—which is an independent assumption in Part I—is here a consequence of the partition geometry: the cosmological horizon has finite area $A = 4\pi c^2/H^2$, and all visible-hidden coupling

passes through this boundary. The number of modes that participate in the trace-out is bounded by A/ϵ^2, which is finite for any $\epsilon > 0$. The effective stochastic process therefore lives on a finite-dimensional configuration space, as Barandes' correspondence requires, without any assumption that the total phase space beyond the horizon is itself finite.

The theorem of Part I applies: the embedded observer's reduced description of the visible sector is P-indivisible, and by Barandes' stochastic-quantum correspondence [10, 11], mathematically equivalent to unitarily evolving quantum mechanics with an action scale $\hbar$ determined by the local structural properties of the partition boundary.

5. Derivation of the Emergent Action Scale ($\hbar$)

5.1 The Classical Horizon Temperature

The theorem guarantees that $\hbar$ exists but does not compute its value. Partition-relativity (§1.4) constrains $\hbar$ to depend only on quantities intrinsic to the partition. The required thermodynamic input is model-independent: it follows from Jacobson's thermodynamic derivation of Einstein's field equations [16].

Jacobson (1995) demonstrated that applying the thermodynamic relation $dE = T\,dS$ to local causal horizons yields Einstein's field equations. The same identity relates mass-energy to area via:

$$dE = \frac{c^2\kappa}{8\pi G}\,dA \tag{21}$$

The classical entropy density on the horizon is $\eta = 1/\epsilon^2$—one degree of freedom per minimal cell (finiteness condition). Therefore $dS = dA/\epsilon^2$, and equating $dE = T_{\text{cl}}\,dS$ fixes the classical temperature:

$$T_{\text{cl}} = \frac{c^2\epsilon^2\kappa}{8\pi G k_B} \tag{22}$$

For the de Sitter horizon, the surface gravity is $\kappa = cH$, giving:

$$k_B T_{\text{cl}} = \frac{c^3\epsilon^2 H}{8\pi G} \tag{23}$$

This temperature depends on G, c, H, ϵ, and k_B—no $\hbar$. It is the temperature of the hidden sector as a classical thermal reservoir, derived from the horizon geometry without reference to quantum mechanics.

5.2 The Emergent Action Scale

The stochastic-quantum correspondence maps the P-indivisible transition matrices to a unitary evolution $U(t)$, but leaves an enormous gauge freedom: the transition matrix constrains only $|U_{ij}|^2$, leaving the quantum phases—and hence the energy eigenvalues

and the overall action scale—undetermined. This is the Schur–Hadamard gauge freedom inherent in the stochastic-to-quantum map. Partition-relativity (§1.4) constrains the resolution: the emergent quantum description is uniquely determined by the partition, so the action scale $\hbar$ can depend only on quantities intrinsic to the partition—the geometric and thermodynamic properties of the boundary between visible and hidden sectors.

The determination of $\hbar$ proceeds in four steps. Steps 1–3 establish the scaling; Step 4 fixes the prefactor.

Step 1: Uniqueness from partition-relativity. The lemma of §1.4 establishes that the emergent description is not merely *constrained* by the partition but *uniquely determined* by it. The three inputs—the dynamics ϕ_t (fixed by Axiom 1), the Liouville measure μ (fixed by Axiom 3), and the partition $\Gamma_V \times \Gamma_H$ (fixed by Axiom 2)—yield a single stochastic process, which maps to a single quantum description up to unitary equivalence. The action scale $\hbar$ is therefore not a free parameter to be fitted but a derived quantity.

Step 2: The action scale is structural, not volumetric. Among the quantities intrinsic to the cosmological partition, some characterize the *structure* of the coupling across the boundary—the discreteness scale, the gravitational coupling, the speed of light—while others characterize the *volume* of the hidden sector—the horizon area A, the number of modes $N_{\text{modes}} = A/\epsilon^2$, or equivalently the Hubble parameter H through $A = 4\pi c^2/H^2$.

The distinction is analogous to a classical thermal reservoir: the temperature of a heat bath determines the Boltzmann factor $e^{-E/k_B T}$ regardless of the bath's total size; adding more degrees of freedom at the same temperature changes the heat capacity but not the thermal fluctuation spectrum. If $\hbar$ depended on H through volumetric factors, observers at different cosmological epochs would obtain different action scales—different quantum mechanics entirely.

This argument is made precise by the following factorization result, combined with the D-gauge completeness theorem of §5.3.

Lemma 5.1 (Volume-independence of transition probabilities)**.** *Let $(\mathcal{C}_V, \mathcal{C}_H, \varphi_t)$ be a deterministic system with transition matrix $T_{ij}(t)$ on $\mathcal{C}_V$. Decompose $\mathcal{C}_H = \mathcal{C}_B \times \mathcal{C}_D$ into boundary modes $\mathcal{C}_B$ and deep modes $\mathcal{C}_D$. If the visible-sector transition is independent of the deep modes on timescales up to t_*—that is, $\pi_V(\varphi_t(x, h_B, h_D)) = \pi_V(\varphi_t(x, h_B, h'_D))$ for all h_D, h'_D and all $t \le t_*$—then $T_{ij}(t) = T^{(B)}_{ij}(t)$ for all $t \le t_*$, where $T^{(B)}$ is computed by marginalizing over $\mathcal{C}_B$ alone.*

Proof. For $t \le t_*$:

$$T_{ij}(t) = \frac{1}{|\mathcal{C}_B||\mathcal{C}_D|} \sum_{h_B, h_D} \mathbf{1}[\pi_V(\varphi_t(x_i, h_B, h_D)) = x_j] \tag{24}$$

By hypothesis, the indicator function is independent of h_D, so the sum over $\mathcal{C}_D$ contributes a factor of $|\mathcal{C}_D|$ that cancels the $1/|\mathcal{C}_D|$ prefactor:

$$T_{ij}(t) = \frac{1}{|\mathcal{C}_B|} \sum_{h_B} \mathbf{1}[\pi_V(\varphi_t(x_i, h_B)) = x_j] = T_{ij}^{(B)}(t) \tag{25}$$

□

Step 3: Dimensional determination. With volumetric quantities excluded, the partition-intrinsic quantities that can appear in $\hbar$ are restricted to those characterizing the local structure of the boundary: c (defines the causal boundary), G (enters through Jacobson's identity relating geometry to energy), and ϵ (sets the phase-space grain). The unique combination with dimensions of action is $c^3\epsilon^2/G$, giving:

$$\hbar = \beta \frac{c^3\epsilon^2}{G} \tag{26}$$

where β is a dimensionless prefactor.

The status of the surface gravity. The surface gravity κ is a local geometric property of the horizon and could in principle contribute to $\hbar$ as a structural quantity. Dimensional analysis settles the question: the most general combination of c, G, ϵ, and κ with dimensions of action is $(c^3\epsilon^2/G) \cdot (\kappa\epsilon/c^2)^e$ for any exponent e. For the de Sitter horizon, $\kappa\epsilon/c^2 = H\epsilon/c \sim Hl_p/c \sim 10^{-61}$. Any non-zero power of e gives a correction suppressed by at least 61 orders of magnitude. The leading-order result $\hbar \propto c^3\epsilon^2/G$ is therefore robust.

Step 4: Thermal self-consistency fixes the prefactor. The derivation so far has produced two independent expressions for the thermal state of the partition boundary, one from each level of description.

Before the trace-out (§5.1), the classical horizon temperature is derived from Jacobson's thermodynamic identity and the entropy density $\eta = 1/\epsilon^2$:

$$k_B T_{\mathrm{cl}} = \frac{c^3\epsilon^2\kappa}{8\pi G} \tag{27}$$

This expression involves only classical quantities: G, c, ϵ, κ, k_B. No $\hbar$ appears.

After the trace-out, the emergent quantum mechanics is established by Part I with an action scale $\hbar$ whose scaling is $\hbar = \beta c^3\epsilon^2/G$ (Step 3) but whose prefactor β is not yet fixed. Quantum field theory on curved spacetime—which is now a *consequence* of the emergent description, not an independent input—predicts the Gibbons–Hawking temperature [17] for any horizon with surface gravity κ:

$$k_B T_{\mathrm{GH}} = \frac{\hbar\kappa}{2\pi c} \tag{28}$$

These two temperatures describe the same physical boundary. Self-consistency requires $T_{\rm cl} = T_{\rm GH}$. For the de Sitter horizon ($\kappa = cH$):

$$\frac{c^3\epsilon^2 H}{8\pi G} = \frac{\hbar H}{2\pi} \tag{29}$$

The Hubble parameter H cancels—confirming the structural/volumetric point. Solving for $\hbar$:

$$\boxed{\hbar = \frac{c^3\epsilon^2}{4G}} \tag{30}$$

The prefactor is $\beta = 1/4$. No free parameter remains.

Why this is not circular. The argument uses the Gibbons–Hawking temperature, which is a result of quantum field theory on curved spacetime—the very theory whose emergence the theorem establishes. The logic is sequential, not circular:

1. Part I establishes that the emergent description *is* quantum mechanics, with *some* action scale $\hbar$ (existence, not value).
2. The emergent QM, applied to the de Sitter horizon, predicts $T_{\rm GH}(\hbar) = \hbar H/(2\pi k_B)$. This is a *consequence* of the emergent theory, expressed in terms of the unknown $\hbar$.
3. The classical temperature $T_{\rm cl}$ is derived independently in §5.1, with no $\hbar$.
4. Matching the two—requiring that the emergent description be consistent with the conditions that produced it—gives one equation in one unknown.

This is a *gap equation*: the same logical structure as self-consistent mean-field calculations in condensed matter physics, where the order parameter is determined by requiring the emergent collective description to reproduce the microscopic conditions that generated it.

5.3 Gauge-Fixing, the Dimensional Obstruction, and the Role of Thermal Self-Consistency

Theorem 5.2 (D-gauge completeness)**.** *Let $U(t) = e^{-iHt}$ be a unitary propagator on an n-dimensional Hilbert space with a fixed configuration basis $\{|i\rangle\}$, where H has non-degenerate eigenvalues $\{E_a\}$ and non-vanishing overlaps $|\langle i|a\rangle| > 0$ for all i, a. If $U'(t) = e^{-iH't}$ satisfies $|U'_{ij}(t)|^2 = |U_{ij}(t)|^2$ for all i, j and all t, then $H' = DHD^\dagger$ for a time-independent diagonal unitary $D = \mathrm{diag}(e^{i\alpha_1}, \ldots, e^{i\alpha_n})$.*

Proof. Write $c_{ia} = \langle i|a\rangle$. Expanding in the energy eigenbasis:

$$|U_{ij}(t)|^2 = \sum_{a,b} c_{ia}\, c^*_{ja}\, c^*_{ib}\, c_{jb}\, e^{-i(E_a - E_b)t} \tag{31}$$

This is a Fourier series with frequencies $\omega_{ab} = E_a - E_b$. Non-degeneracy ensures the frequencies are distinct, so the Fourier coefficients $c_{ia}c^*_{ja}c^*_{ib}c_{jb}$ are uniquely determined. Matching these for U' requires $c'_{ia}c'^*_{ja}c'^*_{ib}c'_{jb} = c_{ia}c^*_{ja}c^*_{ib}c_{jb}$ for all i, j, a, b. Setting $a = b$ and using $|c_{ia}| > 0$: the moduli $|c'_{ia}| = |c_{ia}|$ are fully determined. Writing $c'_{ia} = c_{ia}\, e^{i\delta_{ia}}$ and substituting into the general constraint:

$$e^{i(\delta_{ia}-\delta_{ja}+\delta_{jb}-\delta_{ib})} = 1 \quad \text{for all } i, j, a, b \tag{32}$$

The general solution is $\delta_{ia} = \alpha_i + \beta_a$ for arbitrary phases $\{\alpha_i\}$, $\{\beta_a\}$. The shift $c'_{ia} = c_{ia}\, e^{i(\alpha_i+\beta_a)}$ gives $|a'\rangle = e^{i\beta_a} D|a\rangle$ with $D = \mathrm{diag}(e^{i\alpha_i})$, so:

$$H' = \sum_a E_a |a'\rangle\langle a'| = DHD^\dagger \tag{33}$$

The β_a cancel in the projectors and contribute only unobservable eigenstate rephasing. □

The D-gauge is physically trivial: $U \to DUD^\dagger$ corresponds to $|i\rangle \to e^{i\alpha_i}|i\rangle$, a time-independent basis rephasing that leaves all transition probabilities, energy-level spacings, interference patterns, and oscillation frequencies invariant.

Lemma 5.3 (Cross-interval matching)**.** *Let $U(t_3, t_2)$ and $U(t_2, t_1)$ be propagators with all $|U_{ij}| > 0$, each determined up to D-gauge within its interval. Then the cross-interval transition probabilities $T_{ik}(t_3, t_1) = |\sum_j U_{ij}(t_3, t_2)U_{jk}(t_2, t_1)|^2$ fix the relative D-gauge between intervals up to a single global phase.*

Proof. Independent D-gauges introduce a phase mismatch $\phi_j = \alpha^{(1)}_j - \alpha^{(2)}_j$ at the junction. The requirement $|\sum_j e^{i\phi_j} U_{ij}U_{jk}|^2 = |\sum_j U_{ij}U_{jk}|^2$ for all i, k yields, from the cross-terms:

$$\cos(\phi_j - \phi_l + \theta_{ijlk}) = \cos(\theta_{ijlk}) \quad \text{for all } i, j, l, k \tag{34}$$

where $\theta_{ijlk} = \arg(U_{ij}\bar{U}_{il}U_{lk}\bar{U}_{jk})$. Since θ_{ijlk} varies with (i, k) under the non-degeneracy hypothesis, the only solution is $\phi_j - \phi_l = 0$ for all j, l: the mismatch reduces to a single global phase $\phi_j = \phi$. □

The dimensional obstruction and the role of Step 4. The preceding results resolve all *dimensionless* ambiguity but leave $\hbar$ untouched—and this is a structural feature, not a gap. The unitary $U(t)$ is dimensionless; $\hbar$ enters only when defining $\hat{H} = i\hbar\, \partial_t U \cdot U^\dagger$, converting dimensionless rates into energy units. Different values of $\hbar$ yield different energies but the *same* $U(t)$ and transition probabilities. No quantity of dimensionless probability data can fix a dimensionful constant.

Step 4 provides the necessary and minimal dimensionful input: the thermal self-consistency condition $T_{\mathrm{cl}} = T_{\mathrm{GH}}(\hbar)$ connects the dimensionless oscillation frequencies

to a physical temperature derived from the classical partition geometry (§5.1). This is not a crutch awaiting replacement but the unique completion of the dimensionless-to-dimensionful bridge.

6. Self-Consistency and the Discreteness Scale

Rearranging $\hbar = c^3\epsilon^2/(4G)$:

$$\epsilon^2 = \frac{4\hbar G}{c^3} = 4\,l_p^2 \tag{35}$$

The framework predicts $\epsilon = 2\,l_p$ exactly. The geometric discreteness scale is twice the Planck length—a specific quantitative prediction, not an order-of-magnitude estimate.

This value is necessary for self-consistency. If $\epsilon^2 \ll l_p^2$, sub-Planckian subcells would be dynamically active yet unresolvable by the emergent description, creating a second trace-out with its own noise intensity and breaking the single-valuedness of $\hbar$. If $\epsilon^2 \gg l_p^2$, the emergent description would assign distinct quantum states to configurations that are physically identical in the substratum. The regime $\epsilon = \mathcal{O}(l_p)$ is the only self-consistent one; thermal self-consistency selects $\epsilon = 2\,l_p$ within it.

With $\epsilon = 2\,l_p$, the number of independent modes on the cosmological horizon is:

$$N_{\text{modes}} = \frac{A}{\epsilon^2} = \frac{A}{4\,l_p^2} = S_{\text{dS}} \tag{36}$$

where $S_{\text{dS}} = A/(4\,l_p^2)$ is the Bekenstein–Hawking entropy of the de Sitter horizon. The factor of 1/4 in the Bekenstein–Hawking formula—historically introduced as a proportionality constant—is here derived as a consequence of the partition geometry and the thermal self-consistency condition.

7. Dissolution of the Cosmological Constant Problem

7.1 The Two Levels of Description

The cosmological constant problem [2, 3, 18]—the 10^{122}-fold discrepancy between the QFT prediction for vacuum energy density and the observed value—is often called the worst prediction in physics. Its resolution turns on recognizing that the classical substratum and the emergent quantum description give different—and independently self-consistent—answers to the same question.

From finite-dimensional QM to QFT. The theorem of Part I delivers quantum mechanics on a finite-dimensional Hilbert space. The connection to quantum field theory proceeds through three standard steps. First, the finiteness condition discretizes the visible sector into $N = A/\epsilon^2$ cells of area ϵ^2, each coupled to the hidden sector through

the partition boundary. Second, the causal structure of the classical substratum restricts which cells interact on any given timescale, and the emergent quantum description inherits this locality. Third, spatial locality implies the emergent Hilbert space decomposes as $\mathcal{H} = \bigotimes_k \mathcal{H}_k$—a lattice-regularized QFT with a built-in ultraviolet cutoff at $\epsilon = 2\,l_p$.

The classical substratum (what geometric measurements probe): The cosmological horizon has a classical thermal equilibrium. By the Friedmann equation, $H^2 = (8\pi G/3)\rho$, the horizon area adjusts self-consistently so that the mean energy density matches the critical density $\rho_{\text{crit}} = 3H^2c^2/(8\pi G)$. There is no zero-point energy and no discrepancy. The total vacuum energy density is at the critical scale $\rho \sim H^2/G \sim 10^{-9}$ J/m^3—precisely what is observed.

The emergent QFT (what local quantum measurements probe): Quantum field theory assigns a zero-point energy of $\frac{1}{2}\hbar\omega$ per mode. Summing to the Planck cutoff yields $\rho_{\text{QFT}} \sim M_{\text{Pl}}^4 \sim 10^{113}$ J/m^3.

7.2 Why Gravity Sees the Classical Value

Spacetime geometry is part of the classical substratum (Axiom 2): the metric tensor is defined on the fundamental phase space and evolves via Einstein's field equations *before* the trace-out that produces quantum mechanics. The stress-energy tensor that sources the Einstein equations is the classical stress-energy of the total microstate, not the expectation value of an emergent quantum operator. The 10^{113} J/m^3 zero-point energy exists in the observer's informational ledger—it is a consequence of re-describing the classical noise as quantum fluctuations—but it does not appear in the stress-energy tensor that governs the geometry.

Ontological commitment and its consequences. The framework makes a specific ontological commitment: classical spacetime geometry is fundamental; quantum mechanics is emergent. This is not a free choice but follows from the logical structure of the derivation. The causal partition that produces quantum mechanics is defined by null geodesics of the metric (§4.1); the metric must therefore exist prior to the partition, and hence prior to the quantum description the partition generates.

7.3 The Structural Origin of the 10^{122} Discrepancy

The 10^{122} factor has a precise structural interpretation. Summing the emergent zero-point energies to the Planck cutoff:

$$\rho_{\text{QFT}} \sim \int_0^{\omega_{\text{Pl}}} \frac{1}{2}\hbar\omega \cdot g(\omega)\, d\omega \sim M_{\text{Pl}}^4 \tag{37}$$

The ratio to the classical result is:

$$\frac{\rho_{\text{QFT}}}{\rho_{\text{classical}}} \sim \frac{M_{\text{Pl}}^4}{H^2/G} = S_{\text{dS}} \tag{38}$$

The discrepancy equals $S_{\text{dS}} = A/(4\,l_p^2)$—the Bekenstein–Hawking entropy of the cosmological horizon, now identified as the number of independent modes on the partition boundary ($N_{\text{modes}} = A/\epsilon^2 = S_{\text{dS}}$). This is the information compression ratio of the quantum description: the number of hidden-sector degrees of freedom that the trace-out compresses into the emergent quantum state. The "worst prediction in physics" is the entropy of the observer's blind spot—a category error, not a fine-tuning failure.

8. Predictions and Tests

8.1 Dark Energy Evolution in Running Vacuum Form

The action scale $\hbar$ depends only on structural properties of the partition boundary and is independent of H (§5.2). But the emergent vacuum energy is a state-level quantity that depends on the hidden sector's volume. The horizon area $A = 4\pi c^2/H^2$ evolves with the Hubble parameter, so the emergent vacuum energy inherits a dependence on H. Expanding $\Lambda_{\text{eff}}(H)$ around the present epoch:

$$\Lambda_{\text{eff}}(H) = \Lambda_0 + \frac{3\nu_{\text{OI}}}{8\pi G}(H^2 - H_0^2) + \mathcal{O}((H^2 - H_0^2)^2) \tag{39}$$

This is precisely the form of the Running Vacuum Model (RVM) [20, 21]. The framework requires $\nu_{\text{OI}} > 0$ (the hidden sector gains capacity as H decreases) and $\nu_{\text{OI}} \ll 1$ (condition C3 requires the hidden sector not be saturated). Its value is determined by the spectral structure of the partition boundary.

The S_{dS} horizon modes span frequencies from $\omega_{\text{IR}} = H$ to $\omega_{\text{UV}} = c/\epsilon$. Massless field modes on a 2-sphere have conformal mode density—equal numbers of independent modes per logarithmic frequency interval (a standard result in field theory on curved backgrounds). The trace-out compression noise (§7) is distributed across these $\mathcal{N}(H) = \ln(c/\epsilon H)$ independent spectral channels, so the gravitating vacuum energy scales as $\rho_\Lambda \propto 1/\mathcal{N}$. This is a non-perturbative result: standard QFT gives corrections suppressed by $(H/M_{\text{Pl}})^2 \sim 10^{-122}$, whereas the spectral response gives $1/\mathcal{N} \sim 1/140$. Normalizing at H_0 and Taylor-expanding:

$$\boxed{\nu_{\text{OI}} = \frac{\Omega_\Lambda}{2\,\ln(c/\epsilon H_0)}} \tag{40}$$

With $\epsilon = 2\,l_p$, $H_0 = 67.4$ km/s/Mpc, and $\Omega_\Lambda = 0.685$: $\mathcal{N}(H_0) \approx 139.6$, giving $\nu_{\text{OI}} = (2.45 \pm 0.03) \times 10^{-3}$. The result is insensitive to H_0 (which enters only inside the logarithm)

and yields an independently testable ratio: $\nu_{\rm OI}/\Omega_\Lambda = 1/(2\,\ln(c/\epsilon H_0)) \approx 0.00358$.

DESI's 2024–2025 data releases [22, 30] report evidence for evolving dark energy at 2.8σ–4.2σ. Direct RVM fits by de Cruz Pérez, Gómez-Valent, and Solà Peracaula [31] find $\nu \sim \mathcal{O}(10^{-3})$ with $\nu = 0$ disfavored at 2.7σ–3.1σ. The framework's prediction lies within the phenomenological range from all major dataset combinations. The outstanding discriminating tests are whether the one-parameter RVM form fits the data as well as or better than the two-parameter $w_0 w_a$ model, and whether the measured ν/Ω_Λ ratio converges to the predicted value.

8.2 Gravitational Wave Echoes

A black hole horizon represents a regime where the causal partition differs from the cosmological one. At a proper distance $\sim \epsilon$ outside a black hole horizon r_h, the local wavelength of an infalling mode reaches ϵ—the discreteness floor. The mode has no kinematically accessible states at shorter wavelengths and must scatter back. The predicted time delay is:

$$\Delta t_{\rm echo} \approx \frac{r_h}{c} \ln\left(\frac{r_h}{2\,l_p}\right) \tag{41}$$

For a $30\,M_\odot$ remnant with $\epsilon = 2\,l_p$, the logarithmic dependence on ϵ makes this prediction insensitive to the exact value: replacing l_p with $2\,l_p$ shifts the delay by $\ln 2 \cdot r_h/c \approx 0.1$ ms, well within the ~ 54 ms timescale [23].

8.3 Conjunction as Discriminant

The conjunction of confirmed dark energy evolution *and* detected GW echoes would uniquely favor an information-theoretic origin over standard QFT-derived RVMs, which predict $\Lambda(H)$ running but have no mechanism for discreteness-scale echoes. Each prediction alone is compatible with other frameworks; their joint confirmation would be distinctive.

9. Discussion

9.1 Interpretive Consequences

A note on "hidden sector" in what follows. The general theorem applies to any partition satisfying the axioms, and the interpretive consequences below are stated in that generality. In the cosmological application of Part II, the hidden sector is specifically the trans-horizon region; the degrees of freedom involved in quantum experiments—photons, electrons, slits, detectors—are all visible-sector objects.

In the double-slit experiment, the particle traverses a single slit at the level of the deterministic substratum—it has a definite trajectory in the total phase space. The interference pattern arises because the observer does not have access to the total phase space. What appears as "wave-like" behavior is the trace-out-induced dynamics responding to a change in the visible-sector geometry.

In Wigner's friend scenarios, the Friend's measurement produces a definite outcome. Wigner, who lacks access to the lab's internal degrees of freedom, must trace them out—producing a superposition assignment that reflects his epistemic deficit, not the Friend's physical state. There is no contradiction because the two descriptions correspond to different partitions.

The Everettian measure problem dissolves because the total system evolves deterministically as a single reality—the Schrödinger equation is a mandatory compression algorithm for an observer who cannot track hidden-sector degrees of freedom, and "branches" are features of the compressed description, not of the underlying dynamics.

9.2 Scope of the Theorem

What the theorem says. The characterization theorem (§3.3) establishes a full equivalence: a stochastic process on a finite configuration space is quantum mechanics if and only if it arises from embedded observation under conditions (C1)–(C3).

What the theorem does not say. It does not identify which physical systems satisfy the conditions. That is an empirical question. It does not derive the numerical value of $\hbar$ for any specific system—only that an action scale set by the trace-out noise necessarily exists.

Partition-relativity: physical implications. *Different* partitions produce *different* reduced descriptions. The equivalence does not say "quantum mechanics is the unique description of reality." It says "quantum mechanics and embedded observation under (C1)–(C3) are the same thing."

9.3 Dependence on Barandes' Correspondence

The theorem rests on Barandes' stochastic-quantum correspondence as its mathematical engine. Barandes [10, 11] proves that any indivisible stochastic process on a finite configuration space of size n can be embedded into a unitarily evolving quantum system on a Hilbert space of dimension $\leq n^3$, and conversely that every quantum system can be understood as such a process. The correspondence is bidirectional and exact. The proof rests on the Stinespring dilation theorem and standard linear algebra; it has been published in a peer-reviewed venue [11] and corroborated by independent follow-up work: Calvo [12] extended it to infinite-dimensional systems, Pimenta [25] provided a complete

geometric characterization of divisible and indivisible dynamics, and Doukas [26] proved several key structural properties.

9.4 The Status of the Discreteness Scale

The discreteness scale ϵ warrants explicit discussion, because in a framework that derives quantum mechanics from classical ingredients, introducing a fundamental length scale may appear to smuggle in a quantum assumption. It does not. The finiteness condition (§1.2) requires that the effective configuration space be finite-dimensional—a condition that can be motivated on purely classical grounds. The result $\epsilon = 2\,l_p$ is a *consequence* of the framework, not an input.

9.5 Observable Vacuum Energy Effects

If the quantum vacuum energy is "compression noise," a natural objection arises: why do vacuum energy effects such as the Casimir force and the Lamb shift produce observable consequences? The answer is that these effects are predictions *within* the emergent quantum description, and that description is empirically exact for all visible-sector physics. The Casimir force arises from differences in zero-point energy between boundary configurations—it depends on energy *differences* between geometries, not on the absolute value of the vacuum energy density.

9.6 Relation to Prior Work

't Hooft's cellular automaton interpretation [1] is the closest predecessor. The present framework agrees that quantum mechanics can emerge from underlying determinism, but differs in three respects: the mechanism (epistemic trace-out vs. information loss), the generality (any system satisfying the axioms vs. particular dynamical rules), and the relationship to Bell's theorem (outcome independence violation vs. superdeterminism).

QBism and relational quantum mechanics treat quantum states as observer-relative. The present framework agrees but provides a structural account of *why*: the quantum description arises from a specific mathematical operation (the trace-out) applied to a specific physical situation (the causal partition).

Jacobson's thermodynamic derivation of Einstein's equations [16] showed that the Einstein field equations follow from thermodynamic reasoning applied to local Rindler horizons. The present framework inverts the direction: where Jacobson derives gravity from thermodynamics, the present framework derives quantum mechanics from the causal structure that gravity creates. The two results are complementary.

9.7 Assumptions, Limitations, and Falsifiability

What would falsify each part independently. The *theorem* would be falsified by a restriction of Barandes' correspondence that excludes the class of processes generated here, or by a demonstration that the finiteness condition cannot be satisfied in the relevant physical regime. The *application* would be falsified by the definitive absence of dark energy evolution in precision cosmological surveys, or by the confirmed absence of gravitational wave echoes at the predicted timescale, while leaving the theorem intact.

10. Conclusion

Two results at different levels of generality have been established.

The general theorem (Part I). Under three axioms (deterministic dynamics, a causal partition, and classical probability) and a finiteness condition on the effective configuration space, the P-indivisibility of the reduced dynamics is proved as a mathematical theorem: any bijective dynamics on a finite system with non-trivial coupling (condition C1) necessarily produces P-indivisible stochastic dynamics in the visible sector. Conditions (C2) and (C3)—inverted timescale separation and sufficient capacity—ensure the physical persistence and observational relevance of this indivisibility. By Barandes' stochastic-quantum correspondence, the embedded observer's reduced description is necessarily quantum mechanics. The Schrödinger equation, the Born rule, and Bell inequality violations are structural consequences. The action scale $\hbar$ is uniquely determined by the local structural properties of the partition boundary. No quantum postulates are required. The characterization theorem (§3.3) establishes that the conditions are not merely sufficient but *necessary*: any P-indivisible process, realized through a deterministic dilation, requires non-trivial coupling (C1), slow-bath memory (C2), and sufficient hidden-sector capacity (C3). Quantum mechanics and embedded observation under (C1)–(C3) are equivalent.

The cosmological application (Part II). The cosmological horizon provides a physical realization where the sharp-partition approximation is exact and all three conditions are satisfied. The theorem yields: (a) the exact value $\hbar = c^3\epsilon^2/(4G)$ from partition-relativity, the structural/volumetric distinction, dimensional analysis, and thermal self-consistency, with $\epsilon = 2\,l_p$ and the Bekenstein–Hawking entropy formula $S = A/(4\,l_p^2)$ recovered as consequences; (b) a fundamental dissolution of the cosmological constant problem, recognizing the 10^{122} discrepancy not as a catastrophic fine-tuning failure, but precisely as S_{dS}—the information compression ratio of the observer's quantum description; and (c) falsifiable predictions including dark energy evolution in RVM form with a computed coefficient $\nu_{\mathrm{OI}} = \Omega_\Lambda/(2\,\ln(c/\epsilon H_0)) = (2.45 \pm 0.03) \times 10^{-3}$, consistent with current observational fits, and gravitational wave echoes.

The universe is not broken. The apparent conflict between its two most successful theories is the information-theoretic cost of observing it from within.

Declaration of AI-Assisted Technologies

During the preparation of this work, the author used **Claude Opus 4.6 (Anthropic)** and **Gemini 3.1 Pro (Google)** to assist in drafting, refining argumentation, and verifying bibliographic details. The author reviewed and edited the content and takes full responsibility for the publication.

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